Access ALITA

ALITA runs online, protected by Cloudflare. The research team logs in with an e-mail code, peer reviewers with an invitation code. No password, no installation.

RESEARCH TEAM

Dedicated access with your e-mail

At `alita.prodigi-marburg.de`. Only the research team's e-mail addresses are enabled.

PEER REVIEW

Separate access with an invitation code

Reviewers log in at `review.prodigi-marburg.de` with the invitation code from the editor. No name or e-mail needed.

RESEARCH TEAM: LOG IN IN THREE STEPS

1

Open ALITA

Open `alita.prodigi-marburg.de` or the access link you were sent, in any current browser.

2

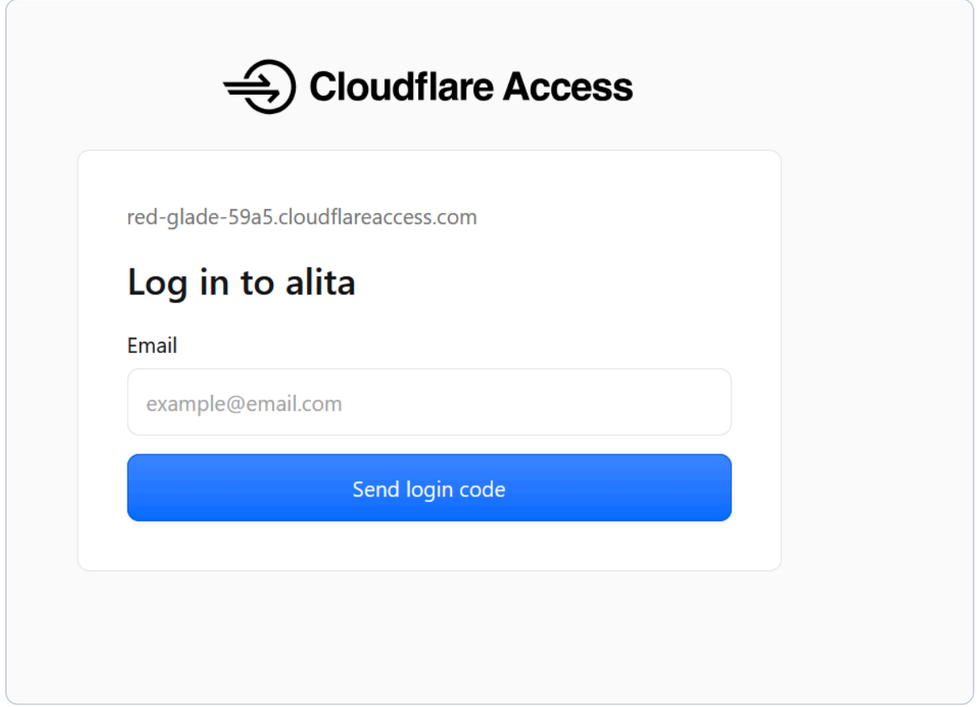
Enter your e-mail

The page **Log in to alita** appears. Enter the address that was enabled for you and click **Send login code**.

3

Enter the code

Cloudflare e-mails you a login code. Enter it on the same page; ALITA opens.



The login page you see in step 2

PEER REVIEWERS: LOG IN WITH THE INVITATION CODE

1

Open the reviewer page

`review.prodigi-marburg.de/review-login.html`

2

Enter the invitation code

Paste the code supplied by the editor into **Invitation code** and click **Continue**.

3

Read and download

Reviewers can read and download all documents but cannot change data.

Keep the invitation code confidential; it is sent separately and not printed here. Cloudflare may process connection information such as your IP address; this is not a guarantee of network anonymity.

WHAT YOU CAN DO ONCE YOU'RE IN

Without an OpenRouter key

Look around:

- Knowledge map, clusters and the 3D view
- Documents list and document details
- Release notes and documentation

With an OpenRouter key with credit

Everything that uses the language model:

- Ask questions in **Document Chat** and **Graph Chat**
- Upload documents with summaries

Remember: each user needs their own key, reviewers too. How to get one → page 2

No login code arriving? (research team)

- Check your spam folder and wait a minute.
- Use exactly the address that was enabled; other addresses get no access.
- Still nothing? Ask the ALITA team to enable your address.

Get an OpenRouter API key

ALITA answers through OpenRouter. You need your own key once; it takes about five minutes.

1 Create an account

Go to `openrouter.ai` and click **Sign in**. Sign up with Google, GitHub or your e-mail address.

2 Add credits

Open `openrouter.ai/settings/credits` and buy credits. OpenRouter works prepaid: requests are paid from this balance.

- Payment by credit card, AliPay or USDC. Card payments carry a 5.5% fee (at least $0.80).
- **Starting with €5 is fine.** At about €0.003 per 10-page paper and well under one cent per question, that lasts for hundreds of questions and uploads.
- Unused credits can be refunded within 24 hours of purchase. Optional: **auto top-up** refills the balance below a threshold you set.

> **Why credits?** ALITA uses a paid model. Free models exist on OpenRouter but are limited to 50 requests per day without purchased credits.

3 Create the key

Open `openrouter.ai/settings/keys` and click **Create Key**.

- **Name:** for example `ALITA`, so you recognise it later.
- **Credit limit:** set a limit (for example $10) or choose no limit. Once the limit is reached, requests with this key are rejected until you raise it.
- **Copy the key right away.** It starts with `sk-or-` and is shown only once.

4 Paste it into ALITA

In the top bar, paste the key into **OpenRouter API-Key** and click **Save**. The status changes from **no key** to **Key …** with its last four characters. The key is stored in this browser; on another device or browser, enter it again.



5 Keep it safe

- Treat the key like a password: don't share it, put it in e-mails or commit it to Git.
- Lost or leaked? Delete it under `openrouter.ai/settings/keys` and create a new one. OpenRouter also e-mails you if it detects an exposed key.
- Check your usage and balance any time under **Activity** and **Credits** in your OpenRouter account.

Checklist

- [ ] Account created
- [ ] Credits added
- [ ] Key copied
- [ ] Saved in ALITA

Fees and limits as stated by OpenRouter, October 2026; they can change. Sources: openrouter.ai/docs/faq · openrouter.ai/docs/api-reference/authentication

Quick Reference

Ask your research documents and get answers with sources you can check.

A · DOCUMENT CHAT
What do the documents say?

B · GRAPH CHAT
How are people, projects and methods connected?

GET STARTED three steps, once

1 Open ALITA

Log in with your e-mail and code (page 1). Pick a view in the top bar: **B** or **A/B**.



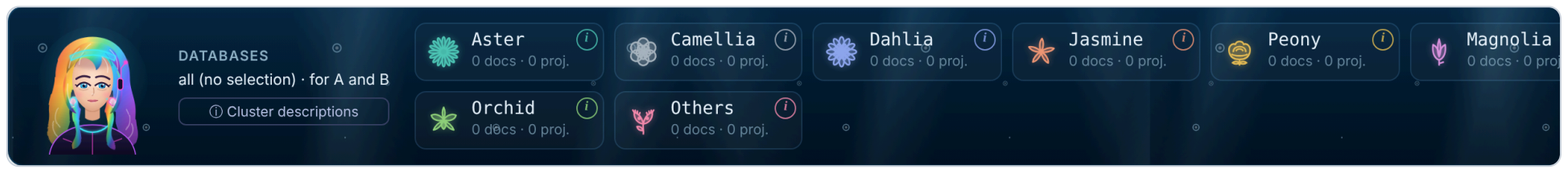
2 Enter your API key

Paste your OpenRouter key (`sk-or-v1-…`) and click **Save**. The browser keeps it. No key yet? How to get one → page 2



3 Choose databases

Click one or more databases. No selection searches all of them. The choice applies to both chats.

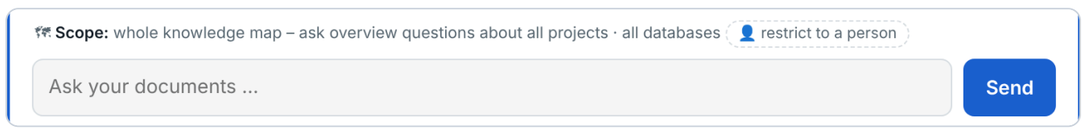


ASK A QUESTION pick the chat that fits the question

ⓘ **Where to find them:** the tab **A/B Interaction Interface** shows both chats side by side. The tab **B Interaction Interface** shows only Graph Chat (B), next to the knowledge map, so **Document Chat (A) is only visible on A/B**.

A Document Chat

RAG (retrieval-augmented generation) searches the **text**: ALITA has cut every document into short passages; it finds the passages closest in meaning to your question and writes the answer from them. Best for content, details, quotes and summaries. Narrow the scope under **Advanced settings** (type, method, cluster, years).

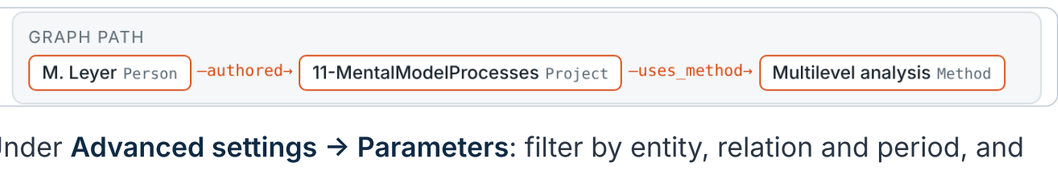


Every claim cites its source as **[1]**, **[2]**. Click a number to see the passage; **Copy** takes the answer with its sources.

> The Mental Model Processes project was written by three authors and submitted to eleven journals [1]. Methodologically, it combines a lab experiment with a multilevel analysis [2].

B Graph Chat

GraphRAG searches the **connections**: on upload, ALITA builds a knowledge graph of people, projects, methods, clusters and document parts, linked by relations such as *authored*, *uses_method* or *cites*. It starts at what your question names and follows these links. Best for relationships: who worked on what, which methods a project uses, who cites whom. The answer shows the path it followed. The graph is based on the SPAR ontologies DoCO, DEO, FaBiO and CiTO (page 6); an ontology for bibliometrics (BIBO) is not implemented.



Under **Advanced settings → Parameters**: filter by entity, relation and period, and set **hops** (1–4) for how far the graph is followed.

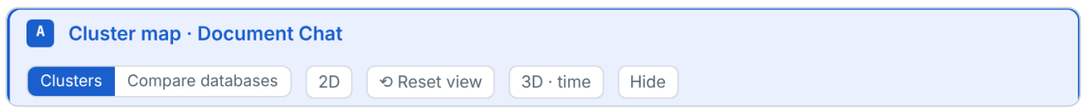
EXAMPLE QUESTIONS

“Who worked with whom on project X?” · more on page 5

EXPLORE AND ADD DATA

Cluster map and 3D view

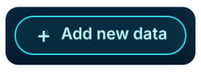
Hover a point to preview, click it to open the document. Click a cluster to limit both chats to it. **Reset view** restores the start view.



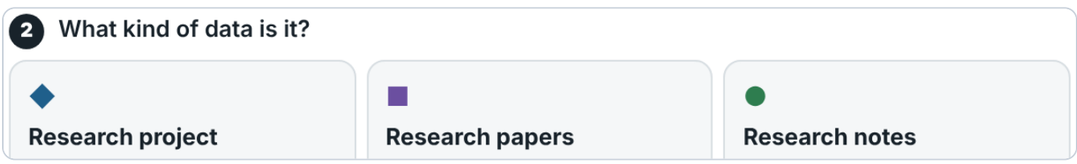
3D · time adds the years as depth: `W` `S` forward/back, `A` `D` sideways, drag to look, `Esc` to close, or **Fly through time**.



Add documents



Click **+ Add new data** (or **Documents → + Upload**). Choose the database, the kind of data, optionally a date, then drop the files and click **Upload**. Projects go in as a ZIP of the folder; tick **Generate summary and methods with the LLM** to have them filled in.



Cost: about **€0.003 per 10-page paper** with the LLM summary (well under one cent; 100 papers ≈ €0.30). Without it, only embeddings: below €0.001. ALITA shows the actual cost after each upload.

Uploading uses your OpenRouter key and credits: set it up → page 2

Estimate: 10 pages ≈ 8,000 tokens of text; summary, section labelling and embeddings ≈ 25,000 tokens; GPT-6 Luna at $0.10 / $0.50 per 1M input / output tokens (OpenRouter, Oct 2026).

Prompt filters

`@db:Magnolia`	only this database
`@type:paper`	only papers (or notes, project)
`@year:2019-2021`	only this period
`@cluster:3`	only this cluster

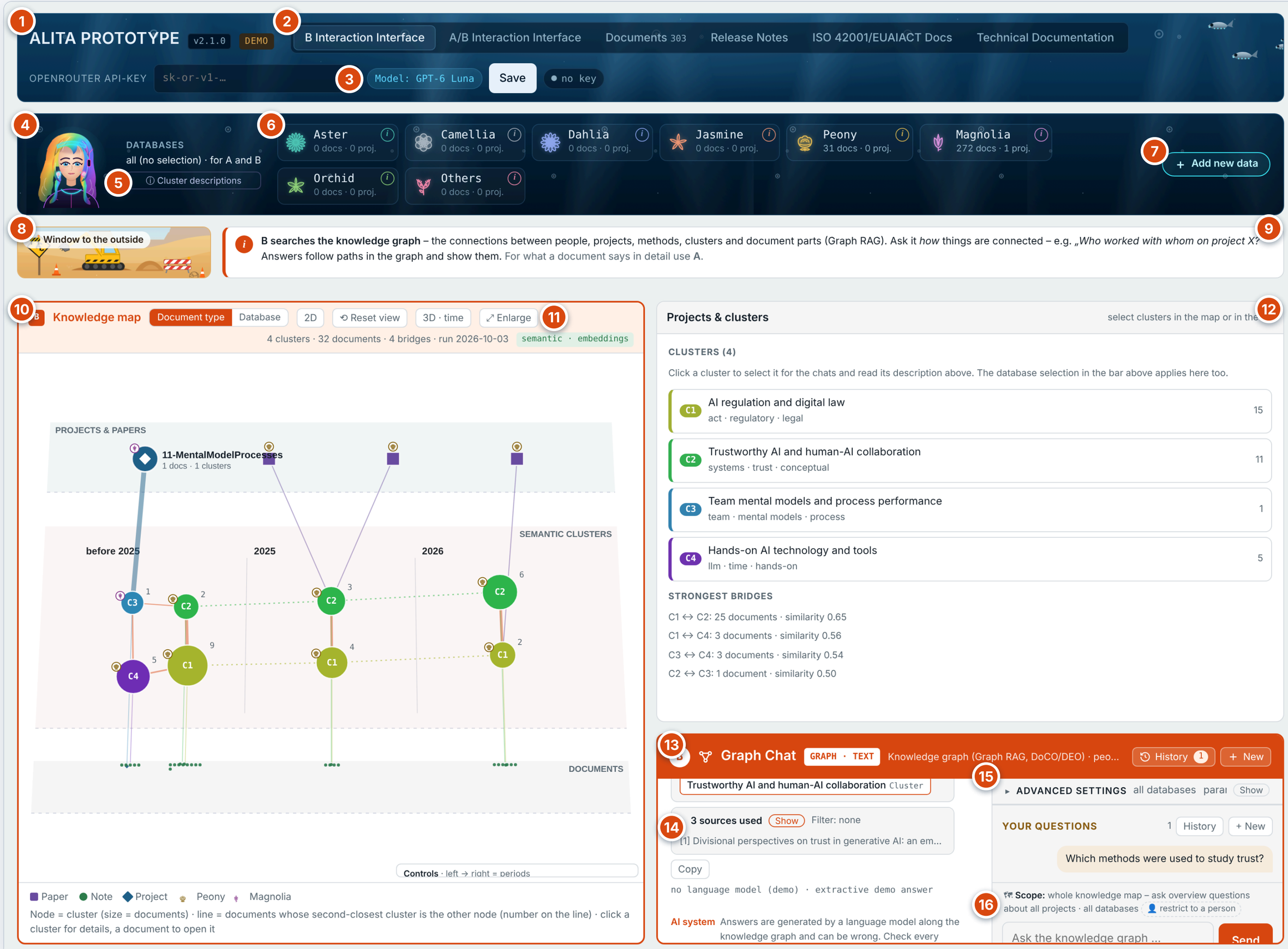
If something looks wrong

No answer	Check the API key is saved (page 2); loosen the filters.
Too narrow	Click **Clear selection** in the database bar.
Off topic	Click **+ New** to start a fresh conversation.
Old chats	**History** reopens them (saved in this browser).

The B Interaction Interface

Knowledge map, projects and Graph Chat on one screen. The numbers match the list below.

B · GRAPH CHAT



TOP BAR AND DATABASES

1 Title and version
Click the version number to open the Release Notes. The DEMO badge appears when the page runs without the backend.

2 Tabs
Switch between **B**, **A/B**, **Documents**, Release Notes and the compliance and technical documentation.

3 API key
Paste your OpenRouter key and click **Save**; the eye shows or hides it. Once saved, the status shows **Key …** with the last four characters. The model is fixed. No key yet? How to get one → page 2

4 ALITA avatar
Follows your cursor. The small **i** buttons on the page say hello and explain the view.

5 Database selection
Shows what is selected; it applies to A and B. **Cluster descriptions** explains all clusters and how the databases share them.

6 Database tiles
Click a tile to select or deselect it; it shows its number of documents and projects. ⓘ opens its contents and cluster descriptions.

7 Add new data
Upload papers, notes or a project ZIP. The selected databases are preset.

8 Window to the outside
Planned external challenge (REQ-20). Opens a concept preview; not active yet.

MAP, PROJECTS AND GRAPH CHAT

9 Info banner
Says what B does: it searches the knowledge graph. For what a document says in detail, use A.

10 Knowledge map
Three layers: projects and papers on top, semantic clusters (C1, C2 …) in the middle, documents below; columns are periods. Circle size = number of documents; lines between clusters = shared documents. Color by **Document type** or **Database**; click a cluster for details, a document to open it.

11 Map controls
2D, **Reset view**, **3D · time** (years as depth) and **Enlarge** to fill the screen.

12 Projects & clusters
Every cluster with its topic, key terms and number of documents; click one to select it for the chats. Below, the strongest bridges between clusters. Project cards and document details open here too.

13 Graph Chat header
History reopens earlier conversations; **+ New** starts a fresh one.

14 Answer
The answer with [n] citations, the **graph path** it followed, the **sources used**, and **Copy**.

15 Advanced settings and your questions
Databases, parameters (entities, relation, period, hops) and the system prompt. Below, your questions; click one to jump to its answer.

16 Scope and question
Shows what the chat searches; **restrict to a person** narrows it. Type your question and click **Send**. Example questions and what works today: use cases → page 5

Use cases for the research proposal

Typical questions, and what the prototype can answer today. ALITA answers from the documents uploaded to it; external literature is outside the prototype.

Current scope of the prototype
This applies to every row below.

- **Within one research project.** Click the project in the map; both chats then answer about that project only.
- **Not yet: between research clusters.** No dedicated comparison of one cluster with another.
- **Not yet: on paper level.** No questions aimed at one single paper or a set of chosen papers.

Status today: Works within one project, possibly once data is uploaded · Partly parts work; scope, input or rules missing · Not in prototype planned for later

USE CASE	TYPICAL QUESTION	WHAT ALITA CAN DO TODAY
YOUR OWN PROJECTS		
1 **Summarise your own knowledge progress across projects**	*"How has my understanding of team mental models developed across my projects since 2015?"*	**Partly – one project at a time** Progress questions are answered from the **knowledge stages** of a project (table of 3, 5 or 10 stages linked to the files), but per project, not across projects. The researchers' own projects are not uploaded yet (to be clarified with Michael).
2 **Identify knowledge leaps and the reasons for them**	*"Where did my thinking change significantly between my early and my recent papers – and why?"*	**Partly – basis to define** Within a project, versions are classified as substantive, rewording, formatting or resubmission. Comparing chosen papers needs paper-level questions (not yet). Open: the basis, e.g. older vs. newer papers of the same authors.
3 **Assess projects and project plans against academic standards**	*"How well does this project plan meet the criteria for a sound research design?"*	**Partly – standards missing** Technically possible. Needed: the project plans (not submitted yet) and the standards, e.g. guidelines as PDF, which can be built into the glossary.
QUESTIONS ABOUT A PROJECT'S DATA AND METHODS		
4 **Explain how the analyses were done**	*"Explain to me how I did the analyses."*	**Works – within one project** Draws on the methods extracted from the files and the analysis documents (scripts, outputs, manuscripts) of the selected project, with sources.
5 **Derive further research questions and loose ends**	*"What further research questions could be derived? Where are the loose ends?"*	**Works – within one project** Answered from the project's limitations, discussion and future-work passages. Not yet across clusters or for one chosen paper.
6 **Find other uses for the data**	*"Can you do anything else with the data?"*	**Works – within one project** Suggestions based on the project's data files, variables and analyses already done. Combining data across projects or clusters is not yet supported.
7 **Uncover research gaps**	*"Which research gaps remain in the 11-MentalModelProcesses project?"*	**Works – tested** Answered from the uploaded documents, with sources. Test results: [to be added].
TOPICS AND IDEAS		
8 **Summarise the state of knowledge on a topic from external sources**	*"What is the current state of research on trust in generative AI?"*	**Not in prototype** External sources were excluded for the initial prototype; answers cover only the uploaded documents. Planned as the **Window to the outside** (REQ-20).
9 **Assess your own research ideas and suggest improvements**	*"Here is my idea: … How strong is it, and how could I improve it?"*	**Works – idea as text** No ideas submitted yet, but none need uploading: paste the idea into the chat. ALITA relates it to the existing documents and cites them.
10 **Support in reaching knowledge leaps**	*"What more should I do with the data? Where do I need to look deeper?"*	**Works – own project needed** Upload the project (**+ Add new data**, ZIP), click it in the map so the chat is limited to it, then ask.

6 work today within one project; some need the researchers' own data or text.

3 work partly: cross-project scope, a comparison basis and academic standards are missing.

1 is outside the prototype: external sources.

Status as of ALITA v2.1.0, October 2026. Example questions are illustrations; answers depend on the uploaded documents.

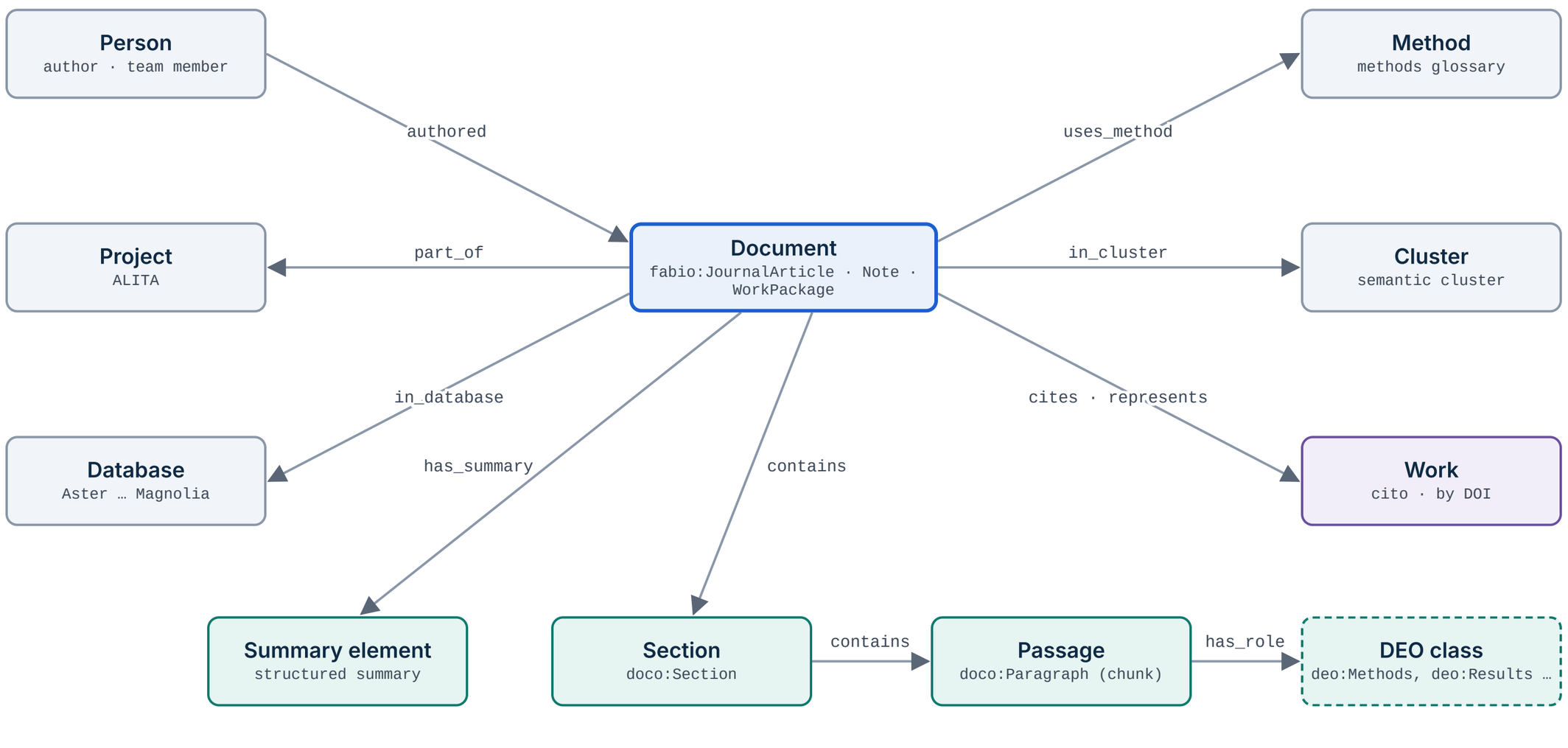
Ontologies behind ALITA

ALITA describes every document with the SPAR ontologies for scholarly publishing and stores the result in a knowledge graph (Neo4j). Graph Chat (B) answers along it; both chats filter by it.

THE ONTOLOGIES

ONTOLOGY	DESCRIBES	HOW ALITA USES IT	EXAMPLES
DEO Discourse Elements Ontology · deo:	The **rhetorical role** of a text part: 31 classes	Every passage, section and summary element gets one or more roles, detected from headings (EN/DE) or by the language model.	Introduction, Methods, Results, Discussion, FutureWork, BibliographicReference
DoCO Document Components Ontology · doco:	The **structure** of a document; imports DEO and the Pattern Ontology	Documents are split into sections and passages in reading order.	Section, Paragraph, Bibliography
Pattern Ontology po:	Pure **containment** structure, the base layer of DoCO	The “contains” links: document → section → passage.	Container, Block
FaBiO FRBR-aligned Bibliographic Ontology · fabio:	The **type** of a publication	Each document is typed by its kind of data.	JournalArticle (paper), Note (notes), WorkPackage (project material)
CiTO Citation Typing Ontology · cito:	**Citations** between works	DOIs in a document's bibliography become “cites” links, so papers in the collection that cite each other are connected.	Document cites Work
ALITA's own project-specific	**Research context** not covered by SPAR	People, projects and teams, databases, the methods glossary and the semantic clusters.	Person, Project, Database, Method, Cluster
BIBO Bibliographic Ontology · bibo:	**Not implemented.** Bibliographic and bibliometric description	ALITA has no bibliometric layer, e.g. citation counts, journals or impact metrics.	–

HOW IT FITS TOGETHER: THE KNOWLEDGE GRAPH



Where you see it in the interface

- **Discourse element** filter in both chats: answer only from, e.g., Methods or Results passages. In Graph Chat it is derived from the question if left empty.
- **Graph path** under each Graph Chat answer, e.g. Person → authored → Project → uses_method → Method.
- **Relation** filter (authored, uses_method, cites, contains, in_cluster) and **hops** in Graph Chat.

Why an ontology

Vector search knows *which document* a passage comes from. The ontology adds *what role it plays* in the argument (method, result, limitation …) and *how documents relate* (authors, methods, citations, clusters). Questions like “Where do the papers discuss limitations?” are then answered along an explicit path instead of by similarity alone.

When each ontology is applied during upload: page 7. Details: tab **Technical Documentation** › Ontology in ALITA. Source: Constantin, Peroni, Pettifer, Shotton, Vitali (2016): The Document Components Ontology (DoCO). Semantic Web 7, 167–181. Ontologies: purl.org/spar/deo · purl.org/spar/doco · purl.org/spar/fabio · purl.org/spar/cito

How documents get into ALITA

The ingestion pipeline, step by step. The ontology is not applied up front: each passage gets its rhetorical role while it is chunked. Clustering runs on the summaries, not on the passages.

Legend: RULE-BASED no language model · LLM GPT-6 Luna, uses the uploader's OpenRouter key · ONTOLOGY SPAR ontologies, see page 6 · EMBEDDING text → vector

1 Extract RULE-BASED

Text and sections with their headings from PDF, Word, Excel …; metadata: type, date, role, version, duplicates, latest version. Methods are tagged by keyword from the methods glossary.

→ **Document record**
metadata, methods; duplicates and files without text stop here (metadata only)

2 Summarise LLM optional · latest version only

A **structured summary** per work: purpose, design/methodology, findings, limitations, originality, plus methods, keywords and a title.

→ **Structured summary**
5 fields per paper, note or project

3 Chunk and assign roles RULE-BASED DoCO · DEO

Passages of 1,200 characters with 200 overlap; the location in the file is kept. Each passage gets its **DEO role** (Methods, Results, Discussion …) from its section heading, in English or German.

→ **Passages with roles**
e.g. “3 Method” → deo:Methods

4 Label the rest LLM DEO

Passages whose heading gives no role are classified by the language model, at most 60 per document to limit cost.

→ **All passages labelled**
ready to be stored

5 Embed EMBEDDING

Every passage becomes a vector, with a short header (type, project, file). Every paper, note and project also gets a **summary vector** from its title and structured summary.

→ **Qdrant · passages**
vector + section + DEO role

→ **Qdrant · documents & projects**
metadata + summary vector

6 Build the knowledge graph DoCO · DEO · FaBiO · CiTO

Document (typed with FaBiO) → sections → passages with their DEO roles; people, project, database, methods, and the DOIs cited in the bibliography (CiTO).

→ **Neo4j · knowledge graph**
nodes and relations for Graph Chat

7 Cluster K-MEANS LLM for names

After an upload (once at the end of a batch): k-means with cosine similarity on the **summary vectors**, one point per paper, note or project; project files inherit their project's cluster. Names and short descriptions by the LLM, map positions by PCA.

→ **Clusters**
written to Qdrant and the graph

WHO USES WHAT

Document Chat (A) · RAG

Searches the **passage vectors** in Qdrant. The discourse-element filter narrows to, e.g., Methods or Results passages.

Graph Chat (B) · GraphRAG

First finds a **path in the graph** (Neo4j), then searches only the passages of the documents on that path.

Knowledge map

Shows the **clusters** from step 7 and their positions; clicking a cluster limits both chats to it.

Defaults from the backend configuration (ALITA v2.1.0): chunk size 1,200 / overlap 200; summaries for the latest version only; at most 60 LLM-labelled passages per document; clustering automatically after each upload.
Code: backend/app/ingest.py, chunking.py, deo.py, graph.py, clustering.py.